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Examination of the Pediatric Patient

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SUMMARY PEARLS

- The pediatric history and physical examination must be tailored to each child and family based off of developmental level, diagnoses, and goals of patient and family.
- The Centers for Disease Control and Prevention's "Learn the Signs. Act Early" program revised its developmental surveillance checklists or milestones in 2022.
- A wide range of developmental tests are available for children based on age, skills, and diagnoses and are useful in approaching assessment in a uniform way.
- Careful observation of the child at play or interacting with others can provide useful information for a pediatric exam.
- An obligatory reflex is never normal, and children should be evaluated for central nervous system disturbance.
- Children's musculoskeletal and neurologic exams are expected to change over time, most rapidly within the first few years of life.
- Children and adolescents often present with musculoskeletal complaints that can have a broad differential diagnosis, and it is important to differentiate between inflammatory and mechanical joint problems.

History

The approach to obtaining the history of children varies based on their age and developmental level. As much as possible, children should be included in the discussion of their health. When the history is obtained from the parent or caregiver, as with very young children, the adult's perception of their child's function and health is often inconsistent with the child's perception. On the initial visit to a pediatric physiatrist, it is helpful to the child and family to summarize the reason for the referral because they may not know why they have been referred. They may have limited understanding of what a pediatric physiatrist is and what they can expect from the visit. A brief description of what the evaluation will entail, the focus on functional assessment, and what the physiatrist can offer the child and family helps set the tone and alleviate anxiety. Asking the child and family what they feel is the underlying cause or problem also assists in evaluating the family's current understanding of the child's medical condition and determining what concerns are important to the family.

The World Health Organization defines adolescence as the phase of life between childhood and adulthood, from ages 10 to 19 years. This marks a time of unique and significant psychological, cognitive, social, and physical developmental changes. With adolescents, it is important to create a safe space where they can freely express themselves. Therefore it may be beneficial to have the parent or caregiver leave the examination area to allow them to provide additional history or ask questions that they may be uncomfortable doing in the presence of family. Throughout the adolescent interview, it is essential to express empathy, promote their

autonomy, empower them to make any needed changes, and maintain confidentiality. However, adolescents should be made aware of specific instances when private information may be disclosed, such as suicidal or homicidal ideation or acts, life-threatening drug misuse, or an eating disorder causing bodily harm.¹⁵

Birth History

A thorough history includes identifying any issues during pregnancy, during labor, and at delivery. Maternal complications during pregnancy, such as seizures, febrile illnesses, hypertension, or hyperglycemia, should be evaluated. Any medications, supplements, alcohol, tobacco, or other substances that the mother used during the pregnancy should be reviewed for potential impact on the fetus. Infants exposed to alcohol or cocaine, for example, often have significant cognitive, behavioral, and motoric impairments. The duration of gestation, presence of multiples, and presentation at birth are also important factors. Premature delivery and multiples are known risk factors for cerebral palsy.^{1,45} The examiner should ask if the mother received prenatal care. If a mother did not receive prenatal care, then she may not have taken additional folate to decrease the risk for spinal dysraphisms.³³ The family should be asked if there was a decrease in fetal movements noted at any time during the pregnancy that could indicate a neuromuscular problem such as spinal muscular atrophy. Ask about accidents or illnesses during pregnancy in the mother or accidents/illness in infancy for the child.

The child's birth weight, length, and Apgar scores should be noted. Higher birth weights with vaginal delivery may lead to a

fractured clavicle and brachial plexus injury. The Apgar score consists of five components, activity, pulse, grimace, appearance, and respiration, each scored from 0 to 2 and recorded at 1 minute, 5 minutes, and 9 minutes, with a score of 7 to 10 considered normal. Postnatal complications, such as hyperbilirubinemia, retinopathy of prematurity, respiratory difficulties, feeding difficulties, and duration of respiratory support, may provide clues to underlying pathology and functional impact. Complications with previous pregnancies, such as stillbirth, miscarriages, or fetal anomalies, should be recorded.

History of Presenting Problem

The physician should determine the onset of the current problem and any associated factors. It should be noted if the symptoms are worsening, static, or improving. It is important to determine which diagnostic tests have been performed and the treatments that have been initiated. Questions regarding the child's temperament and personality can provide insight into the child's readiness to participate in therapies and which approaches may be most helpful. The child's medical history should be reviewed for any similar problems and any significant illnesses, hospitalizations, surgeries, procedures, or previous trauma. Medications should be

reviewed for any possible side effects, particularly on the nervous and/or muscular systems. Allergies should also be reviewed, including any feeding intolerances. A history of early allergies with different formulas may indicate a feeding difficulty rather than a true allergy. Immunization status is important to determine because of the risk for disorders such as tetanus.

Developmental History

Developmental history is one of the most important aspects of pediatric history. Illnesses, injuries, and different disease processes can have a profound impact on the attainment of developmental milestones, with delays noted in gross motor, fine motor, speech and language, and/or psychosocial areas. A thorough understanding of the developmental milestones (Table 2.1) and the age at which the child attained them can assist with diagnosis and treatment protocols.⁵⁶ If the delays are primarily motoric in nature, then a neuromuscular disorder is more likely. If the delays are noted in speech and language skills, then further assessment of the child's hearing is warranted. Parents should be asked if there is a history of recurrent ear infections. Infants with hearing loss start to fall behind after 6 to 8 months of age and may present with decreased babbling.

TABLE 2.1 Developmental Milestones

Age	Gross Motor	Language	Fine Motor	Social
2 months	Moves both arms and legs Holds head up briefly on tummy	Makes sounds other than crying Reacts to loud sounds	Opens hands briefly	Looks at your face Calms when picked up or spoken to
4 months	Good head control when being held Pushes up onto elbows/forearms when on tummy Rolling or rocking side to back	Turns to voice Coos	Grasps toy Moves object to mouth Moves hands to midline	Smiles at face Opens mouth to bottle or breast
6 months	Ability to prop sit Rolls tummy to back	Babbles Makes vowel sounds Laughs	Thumb opposition emerging Reaches with one hand	Recognizes family members Looks at self in mirror
9 months	Sits independently Transitions to sitting May start creeping or crawling ^a	Uses gestures Makes lots of different sounds ("mamama" and "babababa")	Bangs two things together Uses raking grasp or crude pincer grasp Can point	Smiles or laughs at peek-a-boo Is shy around strangers
12 months	Pulls to stand Cruises along furniture Takes steps with hand being held or independently	"Mama" and "Dada" specific Understands "No" Responds to name	Grossly turns pages in a book Puts objects in a container Uses mature pincer grasp	Waves bye-bye Plays patty cake
15 months	Takes steps independently Improved walking skill	Has at least two other words Looks at familiar object when named	Stacks at least two small objects Tries to use objects correctly (cup, book, etc.) Finger feeds well	Copies other children Claps when excited Shows affection (hugs, kisses)
18 months	Stable independent walker. Starting to run Climbs on couch or stairs	Knows 4 to 20 words Follows simple instructions Points to one or more body parts	Builds tower of two to four cubes Throws ball Scribbles	Tries to use utensils Takes off simple clothes
24 months	Kicks ball Runs independently Walks stairs	Uses two-word sentences Knows body parts Is intelligible most of the time (70%) Knows 200 words	Builds tower of six or seven cubes Turns a doorknob Can draw a vertical line	Can put on some clothes Plays with more complex toys (puts toy food on plate) Uses switches, knobs, or buttons on a toy Nods yes/no

TABLE 2.1 Developmental Milestones—cont'd

Age	Gross Motor	Language	Fine Motor	Social
30 months	Jumps off ground with both feet	Says “I,” “me,” or “we” Names things	Twists things like doorknobs or lids	Turns book pages Sometimes plays with other children Follows simple routine instructions
3 years	Balances on one foot Rides a tricycle	Knows first and last name Knows age and sex Uses three-word sentences Can count three objects	Builds tower of 9 to 10 cubes Holds crayon with fingers Copies circle	Dresses Potty trained Separates easily
4 years	Stands on one foot for 5 seconds Catches ball	Uses four- to five-word sentences Counts to 4 Knows colors	Uses scissors Draws circle and square Draws person of two to four parts	Tries to be independent Uses imaginary play Comforts others
5 years	Skiping Stands on one foot for 10 seconds Mature gait Tandem gait Walks backward Hops on one foot	Counts 10 or more objects Uses six- to eight-word sentences Knows coins Knows address	Copies triangle Draws person with a body	Ties shoelaces Sings and dances

^aCrawling has been removed from the Centers for Disease Control and Prevention's most recent developmental milestones checklists.

A discussion of the developmental milestones is also helpful in educating the family regarding what children should be doing and what skills they should be working on. As an example, families are often focused on children walking when they need to take a step back and work on sitting independently or standing balance. It should also be emphasized that there is a wide age range of normal for the attainment of certain skills, and families may notice that their child progresses at different rates within the different areas. Note that in prematurity, corrected age (actual gestational age in weeks – weeks preterm = corrected age) is used in assessing developmental milestones up to 2 years of age. In children with known or severe developmental delays, families often prefer to focus on what their child can do, rather than focus on what they cannot do. Reframing questions to celebrate the accomplishments of these children can help build trust and rapport with their families.

Family History

The family history should include any history of early stroke, early myocardial infarction, peripheral neuropathy, joint or tissue abnormalities, myopathies, and bony abnormalities. Gait abnormalities or developmental delays should be investigated through multiple generations. Certain disease processes, such as myotonic dystrophy, will present in milder forms in previous generations and may be missed until a more severely affected family member presents for medical evaluation. If a genetic disorder is suspected, then the child and family should be referred for genetic testing. Advances in whole exome sequencing and whole genome sequencing have improved the presumptive diagnostic rate of neurodevelopmental disorders and provided guidance in further screening and medication management.³⁸ This can help with planning for future pregnancies and counseling to the extended family. It is important to note, however, that genetic abnormalities due to nucleotide expansions or repeats, such as myotonic dystrophy and Friedreich ataxia, may not be

picked up on whole exome or genome sequencing and may require additional specialized testing.

Social and Educational History

The examiner should inquire about the child's environment, including who lives with the child in the home, the primary and secondary caregivers, the primary language spoken in the home, and the layout of the home. Accessibility is assessed by determining the number of stairs to enter the home, the levels of the home, and where the child's bedroom and bathroom are located. It is important to also evaluate the child's peer group and interactions with peers. Extracurricular activities can give insight into a child's social skills and personality. The child's current educational history should include grade level, presence of an individualized education or 504 plan for additional education supports, any failed or repeated grade levels, and whether the child ever received early intervention services. Early intervention services evaluate and support children, ages 0 to 36 months, who are at risk for delays or have delays in one or more areas (gross motor, fine motor, language).³⁴ It is beneficial to provide families with information on how to obtain and manage special education services among the good resources that are available.^{54,55} A nationally and internationally validated tool to assess adolescent psychosocial history is the HEEADSSS method, which focuses on assessment of the home environment, education and employment, eating, peer-related activities, drugs, sexuality, suicide/depression, and safety from injury and violence (Table 2.2).^{20,21,28} Other useful screening tools include CRAFFT (car, relax, alone, forget, friends, trouble), which is a brief, validated, office-friendly screening test useful in initially assessing adolescent (age 12–26 years) substance use,²⁸ and the Ask Suicide-Screening Questions tool, which is a set of four screening questions to identify suicide risk. A “yes” response to one or more of the questions identified 97% of youth (age 10 to 21 years) at risk

TABLE 2.2
HEEADSSS Assessment

Potential Questions	
Home	Who lives with you? Do you have your own room? Do you get along with the people in your home? Who do you go to if you are upset? Have you ever run away?
Education and Employment	Where do you go to school? Do you like school? What are your favorite classes? Which classes don't you like and why? Are there bullies at school and do they bother you? Have you skipped school? What are your future plans? Do you have a job? Where and what do you do? How many hours do you work?
Eating	What do you typically eat during a day? Has your weight changed recently? How do you feel about your weight and body? Have you ever dieted?
Activities	What do you like to do for fun? Who do you spend your free time with? What outdoor activities do you enjoy? Do you play sports? How much time do you spend on the computer/tablet/cell phone?
Drugs	Do you know anyone who smokes/vapes/used drugs or alcohol? Have you ever tried smoking/vaping/drugs or alcohol? How much? How often? Are you interested in decreasing or stopping your use? Have you had problems related to your alcohol or drug use?
Sexuality	Are you dating anyone currently? What do you like about that person? Have you ever had sex? What protection are you using? Do you like females/males/both? Have you felt unsafe or pressured in any of your relationships?
Suicide/Depression	How do you feel today? Do you ever feel sad or down? Do you feel that people do not understand you? What do you do when you have those feelings? Have you ever thought of harming yourself or others? Do you have a plan? Have you ever tried to end your life?
Safety	Do you feel safe in your home? At school? Do you wear a seatbelt in the car? Do you wear a helmet when riding a bike/scooter/motorcycle? Who can you talk to if you feel unsafe?

Adapted from Goldenring JM, Rosen DS. Getting into adolescent heads: an essential update. *Contemp Pediatr*. 2004;21(1):64-90.

for suicide (Fig. 2.2).²² The CRAFFT tool can be helpful as part of trauma assessment after motor vehicle collisions involving adolescents. Updated versions of the CRAFFT include vaping and edibles for marijuana use and a nicotine checklist for tobacco and nicotine use are available.

Physical Examination

There is no standardized approach to the examination of the pediatric patients.^{2,27} It should be tailored to the age and developmental level of each child. The examination commences with the first introductions. As the examiner is obtaining the history, observation of the child can provide a great deal of information about their social skills, language, motor movements, and personality. Before touching, it is helpful to develop a rapport by playing or talking with each child. A penlight, a badge on a retractable holder, or bubbles are easy tools to engage children and test their visual fixation, reach, grasp, and release at the same time. Knowledge of popular cartoon or movie characters, popular athletes/teams, or singers can be very helpful in engaging children. Very young children are typically most comfortable on their parent's lap during the examination, where they feel safe. Infants are usually tolerant of gentle physical handling by a stranger until about 9 months of age, when stranger anxiety develops. Older children and adolescents can easily be evaluated on the examination table, although autism and other cognitive impairments can require accommodations in the physical exam.

Growth

Height and weight should be plotted and monitored as the child grows. The average full-term newborn measures 50 cm in length;

height increases 50% by age 1 year and doubles by 4 years of age.²⁴ A child's adult height can be estimated by doubling the height at age 2 years. Short stature may be seen with Turner and Down syndromes. Growth may be arrested early because of precocious puberty with premature closure of the growth plates. Precocious puberty, defined as the onset of puberty in females younger than 8 years and males younger than 9 years of age,⁵¹ is commonly seen in cerebral palsy, other brain injuries, and spina bifida.

The average full-term newborn in the United States weighs 3400 g. It is normal for the infant to initially lose weight, but this is regained quickly, and by 5 months of age the weight should have doubled. By 1 year, the child's birth weight should have tripled. Any deviation off the child's expected growth curve should be evaluated closely. Decreases in height velocity or weight loss may be associated with poor nutrition or malabsorption. Significant increases may indicate a pituitary tumor, metabolic disorder, or poor diet. Childhood obesity has reached epidemic proportions in the United States, is considered a health care emergency, and is a particular issue in many children with disabilities. Growth charts are available for specific genetic syndromes (e.g., Turner, Williams, Down syndromes), the cerebral palsy population (based on their gross motor functional classification), and the myelomeningocele population.

In evaluating children, a head circumference should be checked and monitored serially for the first several years. The average head circumference at birth is 35 cm and increases to 47 cm by 1 year of age.²⁴ If the child has macrocephaly, defined as a head circumference greater than 2 standard deviations above the mean, a quick inspection of the parents' head sizes may help to differentiate between a familial trait and hydrocephalus or some type of

TABLE 2.3 Conditions Associated With Microcephaly and Macrocephaly

Microcephaly	Clinical Features
Down syndrome	Brachycephaly, epicanthal folds, short neck, hypotonia, short stature, transverse palmar crease, possible cardiac lesions (ASD, VSD, PDA), intellectual impairment
Trisomy 18 (Edwards syndrome)	Prominent occiput, micrognathia, overriding fingers, short sternum, cardiac lesions (ASD, VSD, PDA), intellectual impairment, high infant mortality
Trisomy 13 (Patau syndrome)	Holoprosencephaly, close-set eyes, coloboma, scalp defects (Fig. 2.1), low-set ears, micrognathia, hypotonia, severe intellectual impairment, transverse palmar crease, polydactyly, cardiac lesions (ASD, VSD, PDA), cleft lip/palate, high infant mortality
Cornelia de Lange syndrome	Generalized hirsutism, prematurity, initial hypertonicity, short stature, confluent eyebrows, long curly eyelashes, long philtrum, low-set ears, micrognathia, speech/intellectual impairments, motor delays, missing fingers
Wolf-Hirschhorn (4p syndrome)	Congenital heart disease, hearing loss, hypotonia, high forehead, short philtrum, prominent glabella, micrognathia, broad nasal bridge, short stature, cleft lip/palate, intellectual impairments, motor delays, seizures
Aicardi-Goutières syndrome	Spasticity, dystonia, abnormal eye movements, cerebral calcification, thrombocytopenia, seizures, progressive loss of function. One of the leukodystrophies.
Angelman syndrome	Severe intellectual impairment, delay in attainment of motor milestones, microbrachycephaly, maxillary hypoplasia, deep-set eyes, blond hair (65%), ataxia, jerky arm movements resembling those of a marionette (100%), seizures
Macrocephaly	
Neurofibromatosis	Type 1: tibial dysplasia, scoliosis, vision impairment, short stature, learning disability, Lisch nodules, café au lait spots, axillary freckling Type 2: hearing loss, balance difficulties, visual impairment, and weakness/numbness caused by compression of spinal nerve roots; symptoms noticed between 18 and 22 years of age
Hydrocephalus	Congenital or acquired: vomiting, sleepiness, seizures, eyes fixed downward, vision impairment, headache, weakness, poor balance/coordination, cognitive impairment, urinary incontinence
Sturge-Weber syndrome	Port-wine birthmark most often on the face, leptomeningeal angioma, glaucoma, hemiparesis, visual impairment, seizures, intellectual impairment in some
Hunter syndrome	Growth deficiency, coarsening of facial features, full lips, macroglossia, contractures of joints, broadening of bones, hepatosplenomegaly, delayed tooth eruption

ASD, Atrial septal defect; PDA, patent ductus arteriosus; VSD, ventricular septal defect.

mass. Hydrocephalus may be present at birth or develop over time. It is commonly seen after closure of a myelomeningocele defect. Microcephaly is a head circumference greater than 2 standard deviations below the mean and indicates that the brain has not fully formed, as in anencephaly, or that growth has been arrested because of some sort of insult, such as anoxia or infection. It may also be seen in certain neurodegenerative disorders.

Inspection

On evaluating the child, initial inspection should assess the child’s general appearance, movements, engagement, and overall health. Close observations of any facial anomalies, joint abnormalities, or asymmetry of stature or sides should be noted. The presence of certain physical abnormalities is linked with some common syndromes (Table 2.3).

Skin examinations may be significant for birthmarks such as port-wine stains, hyperpigmented areas such as café au lait spots (Fig. 2.3), axillary freckling, or acute changes such as the scalded skin appearance of Stevens-Johnson syndrome or the purpura of meningococcal infections. Port-wine stains in the distribution of the first branch of the trigeminal nerve are associated with Sturge-Weber syndrome. A sacral dimple, skin lesions over the spine, and/or a hairy tuft over the lumbar or sacral spine may indicate spina



• **Fig. 2.1** Cutis aplasia or membranous scalp defect associated with Patau syndrome (trisomy 13), although it may be seen in other syndromes as well.

bifida occulta.⁴³ Asymmetric thigh or gluteal folds in an infant are suggestive of hip abnormalities such as dysplasia or dislocation.³² In children with ataxia, telangiectasias are typically present over the flexor surfaces of the elbows and knees. The ash leaf spot in association with seizures and hemiplegia is present in tuberous sclerosis.



Ask Suicide-Screening Questions

Ask the patient:

1. In the past few weeks, have you wished you were dead?	Yes	No
2. In the past few weeks, have you felt that you or your family would be better off if you were dead?	Yes	No
3. In the past week, have you been having thoughts about killing yourself?	Yes	No
4. Have you ever tried to kill yourself?	Yes	No
If yes, how? _____ When? _____		

If the patient answers yes to any of the above, ask the following question:

5. Are you having thoughts of killing yourself right now?	Yes	No
If yes, please describe: _____		



• **Fig. 2.2** Ask Suicide-Screening Questions (ASQ). Suicide risk screening questions. (Courtesy US National Institute of Mental Health, Bethesda.)



• **Fig. 2.3** Café-au-lait spots.

Evaluation of the head and neck should note cranial deformities, facial dysmorphisms, and asymmetries. With the institution of the “Back to Sleep” program in 1992, many children now present to their pediatricians with flattening of the occiput, brachycephaly, or plagiocephaly, which usually resolves when they begin to sit independently.³⁹ Epicanthal folds and palmar simian creases are well-known hallmarks of Down syndrome. Other conditions present with anomalies that include low-set ears, hypertelorism, wide-set eyes, and micrognathia. If the sclera of the eyes has a bluish tinge, then the examiner should consider osteogenesis imperfecta.

Musculoskeletal Assessment

The musculoskeletal assessment includes inspection and palpation of the bones, joints, and muscles. Any asymmetries in muscle mass or limb size should be noted. Lymphedema may present as swelling and increased growth in an upper and/or lower limb. Congenital amputations associated with amniotic band syndrome may include digital, transhumeral, transradial, or transtibial amputations. Apparent

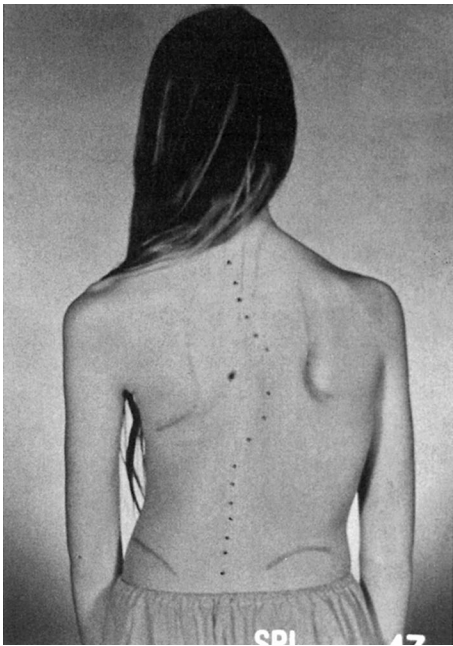


• **Fig. 2.4** Calf pseudohypertrophy.

calf hypertrophy with a doughy feeling on palpation of the muscles is consistent with Duchenne muscular dystrophy (Fig. 2.4). The absence of the pectoral muscle and ipsilateral hand deformities with brachydactyly or syndactyly is described in Poland syndrome (Fig. 2.5). Palpation of the joints evaluates for tenderness, swelling, warmth, and synovial thickening. Both passive and active range of motion of all joints should be evaluated. Joint mobility will change during growth.^{36,46} A full-term infant may lack as much as 25 degrees of elbow extension and 30 degrees of hip extension. A preterm infant, conversely, tends to have increased range of motion because of lower muscle tone. Loss of range of motion may be attributable to joint contracture from arthrogryposis, orthopedic conditions such as Klippel-Feil syndrome, spasticity, pain, inflammatory disorders



• **Fig. 2.5** Absence of the pectoral muscle and ipsilateral hand deformities with brachydactyly or syndactyly is described in Poland syndrome.



• **Fig. 2.6** Evaluation of scoliosis includes an assessment of the spine with the child sitting or standing. This adolescent female has an obvious curve in the standing position and a rotational component.

such as juvenile inflammatory arthritis, infection, or trauma.¹⁶ Connective tissue disorders, such as Marfan syndrome, will result in joint hypermobility and increased skin elasticity. Children who are anxious may not be as easy to examine, and the examiner will have to determine whether there is a volitional component affecting their range of motion.

The back and spine examination should focus on any bony abnormalities and any muscular asymmetries. Children should be examined in both sitting and standing positions because any leg length discrepancy may affect the spine and pelvis examination. Shoulder height, the position of the scapulae, the space between the trunk and the upper limbs, and the height of the pelvis should be evaluated. A rib hump and increased space between the trunk and upper limbs is noted in children with scoliosis (Fig. 2.6). The curve should be evaluated by serial radiographs to determine the severity, flexibility, and progression

TABLE 2.4 Spinal Abnormalities

Spine Abnormality	Clinical Findings
Scoliosis (idiopathic, congenital, neuromuscular)	Curvature of spine on forward bending Rib hump Shoulder asymmetry Pelvic obliquity
Kyphosis (congenital, Scheuermann, neuromuscular)	Abnormal posture increases with flexion
Lordosis (neuromuscular)	Sway back in standing or sitting Coccyx palpable and more protuberant
Spondylolisthesis	Loss of lordosis, reduced range of motion Step-off back deformity Gait abnormalities Transverse abdominal creases



• **Fig. 2.7** Sprengel deformity, a congenital elevation of the scapula, is often seen in association with Klippel-Feil syndrome and cervical spine abnormalities.

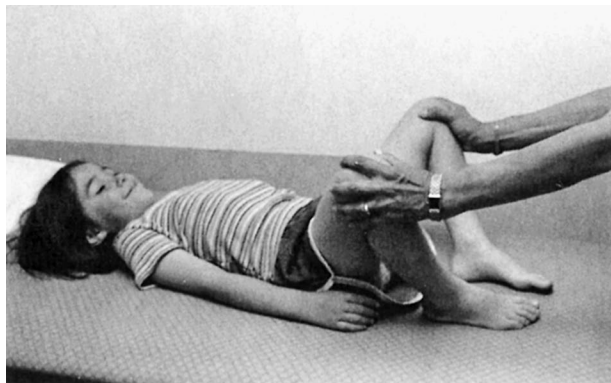
of the curve over time. Idiopathic scoliosis in adolescent females is the most common type of scoliosis, with a right thoracic curve noted.¹³ The rib hump can be further accentuated by having children bend forward. Other spinal abnormalities are listed in Table 2.4. A congenital elevation of the scapula or a Sprengel deformity (Fig. 2.7) is often seen in association with Klippel-Feil syndrome and cervical spine abnormalities.

Evaluation of gait is a common referral for the physiatrist. The lower limb examination may demonstrate range of motion and bony abnormalities. (In addition to evaluating their bodies, children's shoes should be assessed for abnormal wear patterns.) With femoral anteversion, the femur twists between the femoral neck and the femoral condyles, bringing the femoral neck more anterior and causing increased internal rotation at the hip. The normal range of motion for internal rotation is 15 to 25 degrees. Children with femoral anteversion are often "W sitters," and in walking, the patella is noted to be pointed inward. This is most pronounced at 4 to 6 years of age and often self-corrects by 8 to 10 years of

age. In addition to the evaluation of hip rotation, hip abduction should be evaluated (Fig. 2.8). If there is significantly reduced range of motion at the hip or a leg length discrepancy is noted, the hip may be subluxed or dislocated. A Galeazzi sign is present when a thigh length discrepancy is noted with the hips flexed. As the femoral head is pulled superiorly and posteriorly with subluxation or dislocation, the thigh appears shorter in comparison with the normal hip. An anteroposterior pelvis film will delineate the degree of subluxation (Fig. 2.9). Children's knees should be evaluated for stability, mobility, and positioning. When a child presents as bowlegged (genu varus position), this may indicate rickets or Blount disease (Fig. 2.10); however, children have normal physiologic bowing up to 2 years of age. Rickets is caused by vitamin D deficiency and results in softening of the bones' growth plates. The legs will appear bowed and the ankles thickened (Fig. 2.11). Blount disease is more common in the Black population and has been associated with obesity and early walking. It is typically progressive and may need surgical intervention.

Parents are often concerned about the position of their child's feet while walking, whether in-toeing or out-toeing. The most common foot deformity that will appear as in-toeing is

metatarsus adductus (Fig. 2.12) with metatarsal medial deviation. If the foot is flexible, this can often be corrected by a straight-last shoe or wearing the wrong shoe on the affected foot. Inflexible feet will probably require surgical intervention. The most common cause of in-toeing in the toddler is tibial torsion. The distal tibia twists in relation to the proximal tibia.³¹ This can be assessed with the child in prone position by measuring the



• **Fig. 2.8** Examination of the hips should include passive range of motion, such as hip abduction, along with assessment of tone and spasticity.



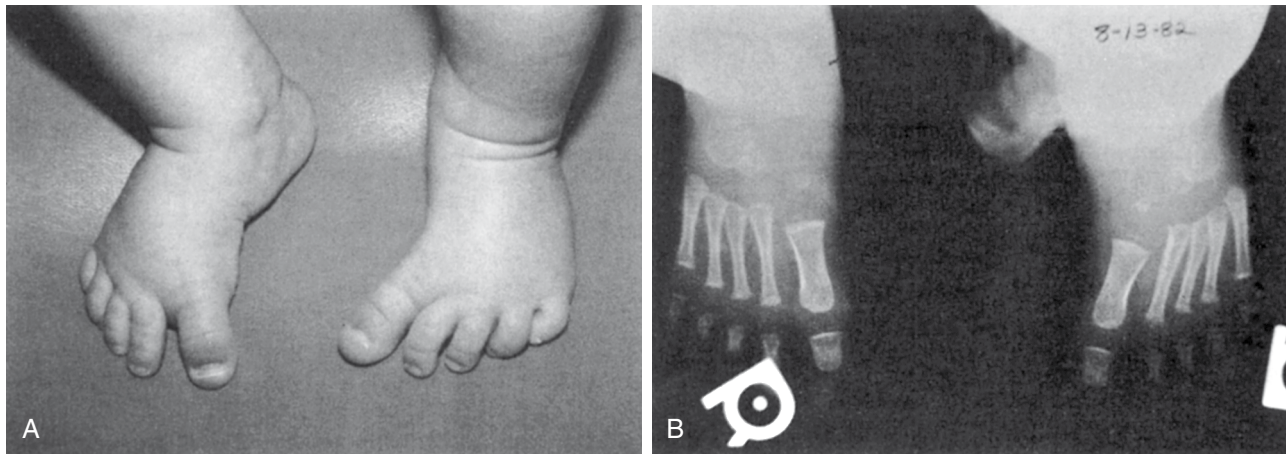
• **Fig. 2.9** An anteroposterior pelvis film delineating the degree of subluxation in hip dislocation; it may be repeated to follow the percentage of migration.



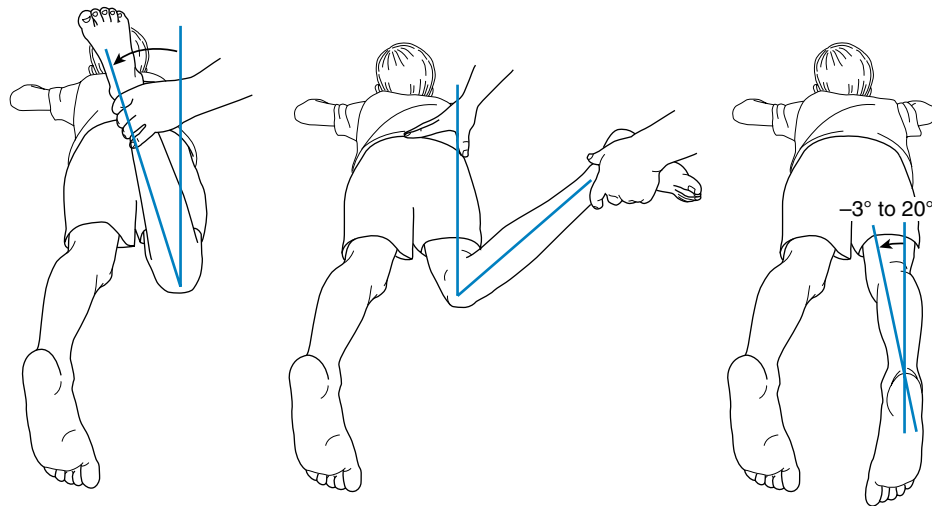
• **Fig. 2.10** Bowlegged (genu varus position) presentation may indicate Blount disease.



• **Fig. 2.11** Softening of the growth plates of a child's bones causes the bowed legs and the thickened ankles seen in rickets.



• **Fig. 2.12** (A) A child with bilateral metatarsus adductus. (B) Radiograph showing medial deviation of the metatarsal bones.

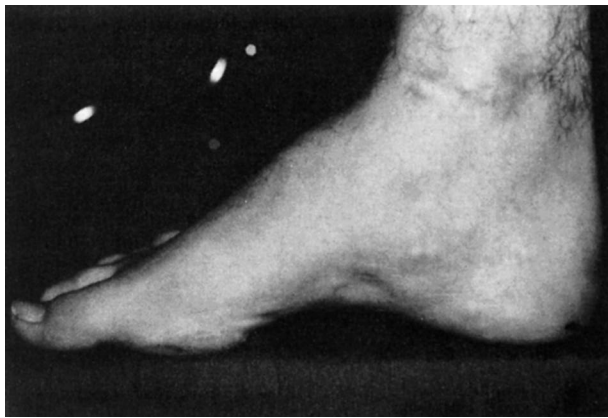


• **Fig. 2.13** Evaluation of a child in the prone position allows assessment of the thigh-foot angle and internal and external rotation of the hip. (Far right) The thigh-foot angle ranges from -3 to $+20$ degrees.

thigh-foot angle (Fig. 2.13). The foot progression angle is assessed with the knee flexed to 90 degrees and then compared with the foot progression angle with the knee fully extended. The angle should range from -10 to $+10$ degrees. In older children, the most common cause for in-toeing is femoral anteversion. Their foot can be normally flat (pes planus) until 3 to 5 years of age.³⁶ A flexible flat foot can be a normal variant into adulthood and does not need intervention; however, if the foot is rigid or painful, this may be the result of a tarsal coalition. The most commonly involved joints are the talocalcaneal and the calcaneonavicular. A rigid flat foot with a rocker-bottom appearance is caused by a congenital vertical talus with dorsal dislocation of the navicular on the talus and has been associated with myelodysplasia and arthrogryposis (Fig. 2.14). Pes cavus is a high-arched foot (Fig. 2.15). When this is associated with sensory loss, foot intrinsic muscle wasting, weakness in the ankle dorsiflexors, and clawing of the toes, further investigation for a sensorimotor neuropathy, such as Charcot-Marie-Tooth disease or other neuromuscular disorders, should be initiated.



• **Fig. 2.14** Rigid flat foot with a rocker bottom caused by a congenital vertical talus with dorsal dislocation of the navicular on the talus associated with myelodysplasia and arthrogryposis.



• **Fig. 2.15** High-arched foot, or pes cavus, is seen in neuromuscular disorders.

Neurologic Assessment

The neurologic assessment evaluates cranial nerve function, sensory function, strength, movement, reflexes, coordination, balance, gait, and cognitive function. Sensory function may be difficult to evaluate in very young children; therefore a complete examination, including an assessment of light touch, proprioception, vibratory, pain, and temperature sensations, may be possible only in older children.⁷ When infants and young children are being assessed, the examiner should use their nonverbal cues, such as looking toward touch stimulus or pulling the limb away when touched. Vibratory toys may be helpful in the evaluation.

Responses to visual stimuli may be attempted with a bright light or an interesting object. Infants will track an object to midline at 1 month of age and from side to side at 3 months of age. Perception of color develops by 2 months of age and depth perception by 3 to 5 months of age. Children should be evaluated for any obvious ocular imbalance. Strabismus is a common finding in children with cerebral palsy. Patching, corrective lenses, and/or surgery is necessary at an early age to improve visual function.³⁰

Auditory evaluation is a part of routine newborn screening. It should be repeated with any child who demonstrates speech and language delays, articulation errors, inattentiveness to sound, history of recurrent ear infections, or history of brain injury.^{6,35} In the infant, hearing can be evaluated by clapping or making a loud noise and watching for a startle (Moro) or blink response. Older children should respond to the rubbing of fingers close to their ears or using toys that make sounds. If abnormalities are noted, the ears should be checked for impaction or infection. If the canals are clear and there are no signs of infection, a comprehensive auditory evaluation is warranted.

Evaluation of primitive reflexes and postural responses is a helpful tool in checking the infant's motor responses. These reflexes (Table 2.5) are controlled at the level of the brainstem and spinal cord.¹⁴ As the central nervous system (CNS) matures, the reflexes are integrated and suppressed between 3 and 6 months of age. Obligatory primitive reflexes, or a reflex that a child cannot move out of, are never normal and indicate a CNS problem. Any asymmetry of reflexes may reflect an underlying stroke or peripheral nerve disorder, such as a brachial plexus injury. Primitive reflexes are replaced by postural reactions (Table 2.6) starting at approximately 2 months of age. These responses allow children to progress with motor development. Asymmetry in these responses

TABLE 2.5 Common Primitive Reflexes and the Period They Typically Disappear

Reflex	Stimulus	Response	Resolves by (Months)
Moro (startle)	Sudden neck extension	Shoulder abduction, elbow and finger extension followed by shoulder adduction and elbow flexion; infant may cry	3 to 4
Rooting	Stroking area around mouth	Head and mouth move toward stimulus	4
Galant reflex	Stroke skin along the side of the spine with the child in prone	Trunk flexes to the side that is stroked	2 to 6
Palmar grasp	Touch palm	Flexion of all fingers	5 to 6
Asymmetric tonic neck	Head turned to side	Arm and leg extend on face side and flex on occipital side (fencer's pose)	6 to 7
Symmetric tonic neck	Neck flexion, neck extension	Arms flex, legs extend; arms extend, legs flex	9 to 11

TABLE 2.6 Postural Reactions and the Period They Typically Occur

Postural Reaction	Stimulus	Response	Age of Emergence
Head righting	Vestibular or visual	Head and face aligned vertically and mouth aligned horizontally	Prone, 2 months; supine, 3 to 4 months
Protective extension	Center of gravity displaced outside base of support in sitting	Abduction of upper limb toward displacement to prevent falling	Sitting anterior, 5 to 7 months; lateral, 6 to 8 months; posterior, 7 to 8 months
Parachute reaction	Center of gravity displaced	Extension of upper limbs outside base of support in standing	Standing, 12 to 14 months toward displacement to prevent falling

is considered abnormal and could indicate hemiplegia or a peripheral nerve injury.

A true manual muscle examination is not very accurate before the age of 5 years, but children should show at least antigravity strength. Strength can be qualitatively assessed by evaluating their ability to

hold the head in midline, sit without support, reach for objects overhead, hold onto objects, creep, stand, and walk. The strength examination is affected by tone and must be interpreted carefully in children with hypertonia resulting from the inability to differentiate between voluntary and involuntary muscle activation. When a child's muscle tone is being evaluated, it is important to remember that the tone will change during development. Premature infants often have lower muscle tone, which should improve with time. If a child is still hypotonic as a toddler or demonstrates motoric delays, then there is likely an underlying neuromuscular abnormality. Hypertonia may be subdivided into spasticity, dystonia, and rigidity.⁴² This is never normal and indicates a CNS disorder. Children with cerebral palsy are often hypotonic as infants and then develop hypertonia within the first year of life. Hypotonia with associated areflexia in a newborn is consistent with spinal muscular atrophy. Acute-onset hypotonia and areflexia in a previously healthy child is suggestive of botulism or acute inflammatory demyelinating polyneuropathy. Tone and reflex exams done on a crying, fussing, or anxious infant or child may yield inaccurate results and should be postponed until the child is calm. This may take several attempts and may be completed with the child on the parent's lap.

Gait assessment requires an understanding of the impact of joint mobility, bony alignment, weakness, and tonal abnormalities on gait. Interventions such as bracing, surgery, and medications will be based on both the examination of muscles at rest and during the functional activity of walking. Children's gait patterns change and progress from the age of 1 to approximately 7 years.²⁶ As their balance and coordination improve, the base of support narrows during the gait cycle. The arms progress from a high guard position to a mature arm swing. The stance phase progresses from a foot flat pattern with an increased foot progression angle to a heel-toe progression with the foot in more neutral alignment. The stride length increases, and the cadence decreases. Common gait deviations are listed in Table 2.7.

Unique factors in the exam of an adolescent patient are the impact of growth spurts along with weight changes on joint and muscle development, being more prone to trauma and sports-related injuries, and the possibility of pain that can stem from physiologic and/or psychosomatic factors. The clinician should screen for conditions such as scoliosis, Osgood-Schlatter disease,

spondylolisthesis, spondylolysis, stress fractures, and Sever disease. Idiopathic scoliosis commonly presents in the adolescent female with a right thoracic curve, although may also be seen in males. Neuromuscular scoliosis may develop or worsen with the adolescent growth spurt. Visual inspection of the spine in sitting, standing, and bending forward at the waist is recommended (Fig. 2.6). Scoliosis should not be painful. Back pain in the adolescent, especially adolescents involved in gymnastics or cheerleading, should be further evaluated for spondylolisthesis or spondylolysis. Complaints of anterior knee pain or heel pain in the adolescent, especially with a growth spurt or involvement in joint loading activities is a common presentation for Osgood-Schlatter disease and Sever disease respectively. In Osgood-Schlatter disease, a tight patellar tendon causes a traction apophysitis of the anterior tibia. A painful bump over the anterior tibial tubercle can be palpated. This is commonly seen in boys ages 10- to 15 involved in running or jumping sports. Sever disease is a traction apophysitis at the insertion of the Achilles tendon on the calcaneus and also commonly seen in athletes involved in running and jumping activities. Screening of the adolescent concussion is a common referral. Protocols for return to play are well established. Please see Chapter 44 on traumatic brain injury for a further discussion of this topic.

When assessing pain in adolescents, it is best practice to not only use standardized pain assessment tools like the visual analog scale and do a comprehensive physical exam but also to observe nonverbal cues such as guarding or change in facial expression on palpation. With adolescent patients, puberty can cause not only physical changes but also mental and emotional changes. It is imperative to look for signs of self-injury or abuse, such as cuts, scratches, or bruises. The clinician should have a higher index of suspicion of self-injury if an adolescent is wearing long sleeves or long pants in hot weather.

Functional Assessment

There are many tools that can be used to evaluate and quantify functional status in the pediatric population. The commonly used Functional Independence Measure (FIM) for adults has been adapted to children (weeFIM) by adjusting for developmental stages. The levels of assistance are the same between the two scales, but the expected FIM level for age differs based on developmental level (e.g., a 12-month-old would be expected to be at weeFIM level 1 [complete dependence] for toileting). This tool has been used successfully to measure progress and outcomes for pediatric inpatient rehabilitation services as well as in the outpatient setting. Developmental skills can be formally assessed with a variety of tools (Table 2.8). The Denver Developmental Screening Test (DDST-II) is useful as a quick screen to determine whether a child is achieving developmental milestones in the areas of gross motor, fine motor, language, and personal social skills.¹⁹ It can be used with children from birth to 6 years of age. The DDST-II is both an observational assessment and an assessment based on parental report. If a child fails the Denver screening, a more formalized assessment with the Bayley Scales of Infant and Toddler Development³ or the Gesell Developmental Schedules¹⁷ is recommended. These tests are typically given by therapists. The Bayley test is appropriate for children from birth to 30 months of age, and the Gesell test is for children from 4 weeks to 6 years of age. These tests are based on direct observation of the child's skills. Their major drawback is that cognitive ability is estimated based on motoric performance. In children with significant physical impairment, the tests often underestimate cognitive abilities.

TABLE 2.7 Gait Abnormalities

Gait	Characteristic(s)	Clinical Association
Spastic	Adducted hips Internal rotation of hips Toe walking	Cerebral palsy
Crouched	Weak quadriceps Weak hip extensors Excessive dorsiflexion Hip or knee contractures	Neuromuscular disease Cerebral palsy
Hemiparetic	Posturing of upper limb Circumduction of hip Inversion of foot	Cerebral palsy Cerebral vascular accident
Waddling (Trendelenburg)	Weakness of hip girdle Wide-based gait	Neuromuscular disease
Ataxic	Coordination problems Poor tandem walking	Cerebellar ataxia Friedreich ataxia

TABLE 2.8 Developmental Evaluation and Screening Tests

Test	Age Range	Scope and Value
Denver Developmental Screening Test ¹⁷	Birth to 6 years	Quick screen for deviations from normal development of normal and near-normal children; pattern of functional deviations guides further evaluation
Bayley Scale of Infant and Toddler Development ³	Birth to 30 months	Separate mental and motor scales; well standardized; heavily weighted with motor-based items, which limits predictive value in children who are physically disabled
Gesell Developmental Schedules ¹⁹	4 weeks to 6 years	Indicator of current developmental level

TABLE 2.9 Perceptual Evaluations

Test	Age Range (Years)	Scope and Value
Beery-Buktenica Developmental Test of Visual-Motor Integration ⁴	2 to 16	Assesses visual motor performance, ability to copy geometric shapes, age equivalence
Bender Visual-Motor Gestalt Test ⁵	5 to adult	Assesses visual motor performance, easy to administer, nine geometric designs

To evaluate visual motor abilities, the Beery-Buktenica Developmental Test of Visual-Motor Integration⁴ and the Bender Visual-Motor Gestalt Test⁵ are most commonly used (Table 2.9). These tests can detect delays in visual perceptual skills and eye-hand coordination from ages 2 years through adulthood. The Rey-Osterrieth Complex Figure test⁹ can further test visual organization and visual memory. It entails the copying of a complex figure followed by delayed recall of the figure. The test may also assist in discerning visual field cuts or areas of visual neglect. The test is appropriate for children 6 years of age to adulthood. There are additional evaluation tools to assess perceptual motor, auditory processing, and tactile functioning.

Cognition and potential for academic achievement can be assessed in preschool and school-age children with several different tests (Table 2.10). These tests evaluate both intellectual and physical abilities and are primarily based on motor function and linguistic ability. For children with physical limitations or language impairments, alternative nonverbal and nonmotor tests are available (Table 2.11). Of note, vocabulary tests have the highest correlation with school success and overall intellectual ability. Academic achievement tests measure a child's proficiency in school subject areas such as reading and mathematics. Table 2.12 lists some

TABLE 2.10 Intellectual Evaluations

Test	Age Range (Years)	Scope and Value
Stanford-Binet Intelligence Scale ⁴⁷	2 to adult	Detailed diagnostic assessment (mental age and IQ); guidelines for hearing, visual, and motor handicaps
Wechsler Preschool and Primary Scale of Intelligence-Revised ⁵⁰	3 to 6.5	Verbal, performance, and full-scale scores; delineates strengths and weaknesses; not appropriate for children with severe developmental delays
Wechsler Intelligence Scale for Children-Revised ⁴⁹	6 to 16	Verbal, performance, and full-scale scores; subtests point to specific areas of strength or dysfunction
Kaufman Assessment Battery for Children ²⁵	2.5 to 12	Measures mental processes independent of the content of acquired knowledge; useful for children from disadvantaged backgrounds

IQ, Intelligence quotient.

TABLE 2.11 Alternative Nonverbal and Motor-Eliminated Tests

Test	Age Range (Years)	Scope and Value
Peabody Picture Vocabulary Test ¹²	2.5 to 18	Effective test of language, especially in children with speech and motor impairments
Leiter International Performance Scale ²⁹	2 to 18	Measures nonverbal problem-solving abilities in children who are deaf and speech and motor impaired
Pictorial Test of Intelligence ¹⁸	3 to 8	Measures intellectual ability of children with multiple impairments; requires receptive language
Raven's Progressive Matrices ⁴⁰	6 to adult	Measures nonverbal intelligence and concept formation

of the more commonly used tests. The scores are reported as grade equivalents. Subsets of these tests can be administered on the inpatient unit to assess a child's current academic level. Such tests are particularly helpful for children who have sustained a brain injury. The information assists with reintegration back into the school setting by establishing a baseline of knowledge from which to proceed. In addition to physical, perceptual, and cognitive skills, it is important to evaluate a child's social and adaptive skills

TABLE 2.12 Academic Achievement Tests

Test	Grade Level or Age Range	Scope and Value
Wide Range Achievement Test—Revised ²³	Kindergarten to 12th grade	Yields academic achievement level in reading, spelling, arithmetic; can measure progress
Woodcock-Johnson Psychoeducational Battery: Test of Achievement ⁵³	3 years to adult	Yields age and grade level, percentiles, and standard scores in reading, mathematics, written language, and general tasks
Peabody Individual Achievement Test ¹¹	Kindergarten to 12th grade	Only pointing response for overview of achievement; useful for handicapped

TABLE 2.13 Social and Adaptive Skills

Test	Age Range	Scope and Value
Vineland Adaptive Behavioral Scale ⁴⁴	1 month to adult	Questionnaire of social competence in communication, socialization, daily living skills, and motor skills; adjusted for disability
American Association of Mental Deficiency Adaptive Behavior Scale ³⁷	3 years to adult	Activities of daily living; adaptive and maladaptive behaviors; assists in program planning

TABLE 2.14 Quality-of-Life Measures

Test	Age Range (Years)	Scope and Value
Pediatric Quality of Life Inventory self-report and parent-proxy questionnaire	2 to 18	Measures health-related quality of life
Child Health Questionnaire self-report and parent-proxy questionnaire	5 to 18	Measures 14 unique physical and psychosocial concepts

and perceived quality of life (Tables 2.13 and 2.14). The Vineland Adaptive Behavior Scale⁴⁴ is appropriate for children with and without disabilities. It measures a child's ability to use skills to function in everyday settings. It is composed of four components: communication, daily living, socialization, and motor skills. It also has a scale for maladaptive behaviors that surveys inappropriate social or behavioral displays. Information is collected from the child, the parent or caregiver, and the school. The Pediatric Qual-

ity of Life Inventory evaluates a child's opinion regarding satisfaction with personal health and wellness.⁴⁸ It measures the child's perceived quality of life in physical, emotional, social, and school functioning.

Disability-specific assessment tools include the Gross Motor Function Measure 66 (GMFM-66),⁴¹ the Manual Ability Classification System (MACS), and the Quality of Upper Extremity Skills Test (QUEST) for cerebral palsy. The GMFM-66 is a subset of the original GMFM-88; it evaluates gross motor functions such as lying, rolling, sitting, crawling, standing, walking, running, and jumping. Individual items are scored on a 4-point scale ranging from 0 (no initiation) to 4 (completion of the task). The MACS and the QUEST both evaluate fine motor skills and hand manipulation. The MACS categorizes children into one of five levels of function ranging from level 1 (handles objects well and easily) to level 5 (unable to handle objects). The QUEST evaluates movement patterns and hand function.¹⁰

Special Considerations for Adolescent Patients

Driving is a symbol of independence for adolescents; however, it is a great responsibility with many risks that must be addressed. Foremost, traffic accidents are the leading cause of death in youth in the United States.⁸ The clinician can provide education on safety and evaluate an adolescent's readiness to drive. The laws governing which medical conditions may prevent someone from driving will vary across states and countries. If there are concerns about an adolescent's ability to drive safely, then the clinician should consider a referral to occupational therapy for a driver's evaluation program.

Transition of adolescent care to an adult provider is essential for improved clinical outcomes and ensures continuity of developmental and age-appropriate care. Clinical guidelines recommend that clinicians provide guidance and support to ensure a planned transition from pediatric to adult health care, beginning at the age of 12 years.⁵² Adolescents with intellectual and developmental disabilities face unique challenges that significantly impact their health outcomes, emotional well-being, and quality of life. Helping the adolescent and family better advocate for their needs is paramount for the clinician. Various tools, from neuropsychological testing to individualized career planning, can help in future planning for adolescents with disabilities. These can help guide the family and clinician in college aspirations, help find appropriate employment, or determine whether the adolescent would do better with prolonged high schooling. A day program is another option and can improve an adolescent's quality of life and socialization along with providing respite for parents or other caregivers. In some situations, it may be best for the parent or another trusted adult to gain guardianship when the adolescent turns 18 years old. Guardianship, in this context, is a court-ordered relationship where one adult (the guardian) is authorized to make decisions for and act on behalf of another adult (the ward) who is unable to make sound decisions for oneself because of a disability. Please note that the court personnel will investigate if a less restrictive alternative to guardianship exists, such as power of attorney or supported decision making. The clinician can aid in this process by assisting with any documentation needed, educating, and providing legal resources.

Summary

The pediatric history and physical examination should be tailored to each child and family. The child's developmental level must be evaluated and rehabilitation approaches selected and instituted

based on that developmental level and the goals for continued development. A wide range of assessment tools are available to assist with determining each child's strengths and weaknesses in all domains.

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